

Requirements

1 Functional Requirements

1. User Management

- The system shall allow customers to access the e-commerce platform through web/mobile interfaces.
- The system shall authenticate users and maintain their authorization context.

2. Product Management

- The system shall allow customers to browse products.
- The system shall maintain product information such as product name, description, price, category, and availability.
- The system shall support product categories and flash-sale products.

3. Cart Management

- The system shall allow customers to add products to their cart.
- The system shall allow customers to update or remove cart items.
- The system shall calculate the cart/order total.

4. Inventory Management

- The system shall maintain available and reserved inventory quantities.
- The system shall check product availability before purchase.
- The system shall update inventory when products are purchased or reserved.
- The system shall support atomic inventory operations to prevent inconsistent stock counts.

5. Inventory Reservation

- The system shall reserve inventory for an order during the purchase process.
- The system shall maintain reservation status and expiration time.
- The system shall prevent multiple customers from reserving the same unavailable inventory.
- Expired reservations shall be detected and released.

- Reservation operations shall support idempotency.

6. Order Management

- The system shall create orders from customer purchases.
- The system shall maintain order status and total amount.
- The system shall process order and reservation state transitions.
- The system shall publish relevant order events such as order confirmation.

7. Payment Management

- The system shall initiate payment requests through an external payment gateway.
- The system shall record payment amount, method, status, and transaction reference.
- The system shall handle successful and failed payment events.
- The system shall maintain payment-related records for an order.

8. Shipment Management

- The system shall create shipment information for confirmed orders.
- The system shall maintain carrier and tracking information.
- The system shall track shipment and delivery status.

9. Notification Management

- The system shall generate notifications for relevant order and payment events.
- The system shall support events such as order confirmation and payment failure.

10. Monitoring and Reconciliation

- The system shall collect metrics, logs, and traces.
 - The system shall support reconciliation between Redis stock counts and database inventory counts.
 - The system shall detect inconsistencies between inventory data stores.
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2 Non-Functional Requirements

1. Performance

- The system should support high-volume product browsing and purchasing.
- Inventory reservation and stock updates should be processed with low latency.

2. Scalability

- The system should scale horizontally to handle increasing numbers of customers and transactions.
- Individual services should be independently scalable.

3. Availability

- The platform should remain available during high traffic and flash-sale periods.
- Failure of an individual service should have minimal impact on other services.

4. Consistency

- Inventory quantities must remain consistent between the cache and persistent database.
- Reservation and order operations must avoid duplicate processing.

5. Reliability

- Payment, order, inventory, and reservation operations should be reliable.
- The system should support retry-safe/idempotent operations.

6. Security

- Authentication and authorization shall be enforced for protected operations.
- Payment information shall be handled through a secure external payment gateway.
- APIs should implement appropriate security controls such as authentication and rate limiting.

7. Observability

- The system should provide centralized metrics, logs, and distributed traces.
- Important business events should be observable through event processing.

8. Maintainability

- The system should use modular services with clearly defined responsibilities.
- Components should follow suitable design principles and patterns to simplify maintenance.

9. Fault Tolerance

- The system should handle failures of external services such as payment and shipping providers.
- Event-based processing should allow failed operations to be retried or reconciled.

10. Data Integrity

- Database constraints and conditional updates should prevent invalid inventory and order states.
- Reservation and payment records should maintain referential integrity.